Run report: XP-20261016 prime-base oracle tuning, run.sh all (2026-10-29T01:29Z)

This report records observations only. The decision is in the study's `README.md`.

What ran

- **Command:** `sh backlog/XP-20261016-prime-base-oracle-tuning/run.sh all research/prime_base_oracle_tuning_20261029` with `RAYON_NUM_THREADS=1` and `PBO_CPU=3` (`taskset -c 3`).

- **Launch:** frozen task card `CARD-XP-20261016-prime-base-oracle-tuning-01` (experiment director → mechanical executor).

- **Timing:** started 01:29:21Z, analysis written 01:56:09Z.

- **Commit and binary:** worktree commit `7df34ba67dc6`; binary source `0e124fb1b207`; ecbench SHA-256 `bf5eec6d…3807d9`.

- **Host:** 4-vCPU Intel Xeon @ 2.10 GHz cloud container, env class `ECBENV3hc4d998f5ae3f` (`host/host.json`). Isolation: one pinned core, shared host. No dedicated-host claim.

- **Stages, in order:**

- host

- identity: replay of frontier P20–P32

- train: 4 sessions + 6 edge-extension sessions at 20 bits

- select

- holdout

- kummer

- audit

- analyse

- **Verify:** `run.sh verify`, run afterwards, wrote `verify/`.

Runs requested and completed

stage	sessions	measured records	verified
identity (replays of frontier P20–P32)	4	240 each	240 each; replays reproduced
train P20 / P24 / P28 / P32	4	232 / 232 / 208 / 208	all
train edge extension (20 bits; mitm3, rfmitm3, rfsub3 at N 91, 128)	6	16 each	all
holdout P20 / P24 / P28 / P32	4	1,008 each (7 arms × 48 targets × 2 rounds, plus warm-ups)	all
kummer P22 / P24	2	160 each	all

No run was exhausted, timed out, errored or ran out of memory. Every audit printed `ok` with 0 problems (`audits/`, `pbo-run.log` lines).

Measured values

All values are *recorded*. They are in the unit S = counted GAE/√r, priced at w = 0, 0.1 and 1 GAE per native operation.

- Holdout means per arm: `analysis/band-holdout.md` and `analysis/table-holdout.md`. The latter also gives 95% intervals of the mean, and median wall time per run, which is not a claim.

- Paired ratios: `analysis/compare-*.json`, 96 pairs each.

- Train cells: `select/band-train-P*.md`. Selection: `select/selection-P*.json`.

Selected cells (train rule: minimum mean S at w = 0.1):

size	selected	incumbent
20	`mitm2-n45`	`mitm2-n32`
24	`mitm2-n115`	`mitm2-n162`
28	`mitm2-n288`	`mitm2-n407`
32	`mitm2-n1024`	`mitm2-n1024`

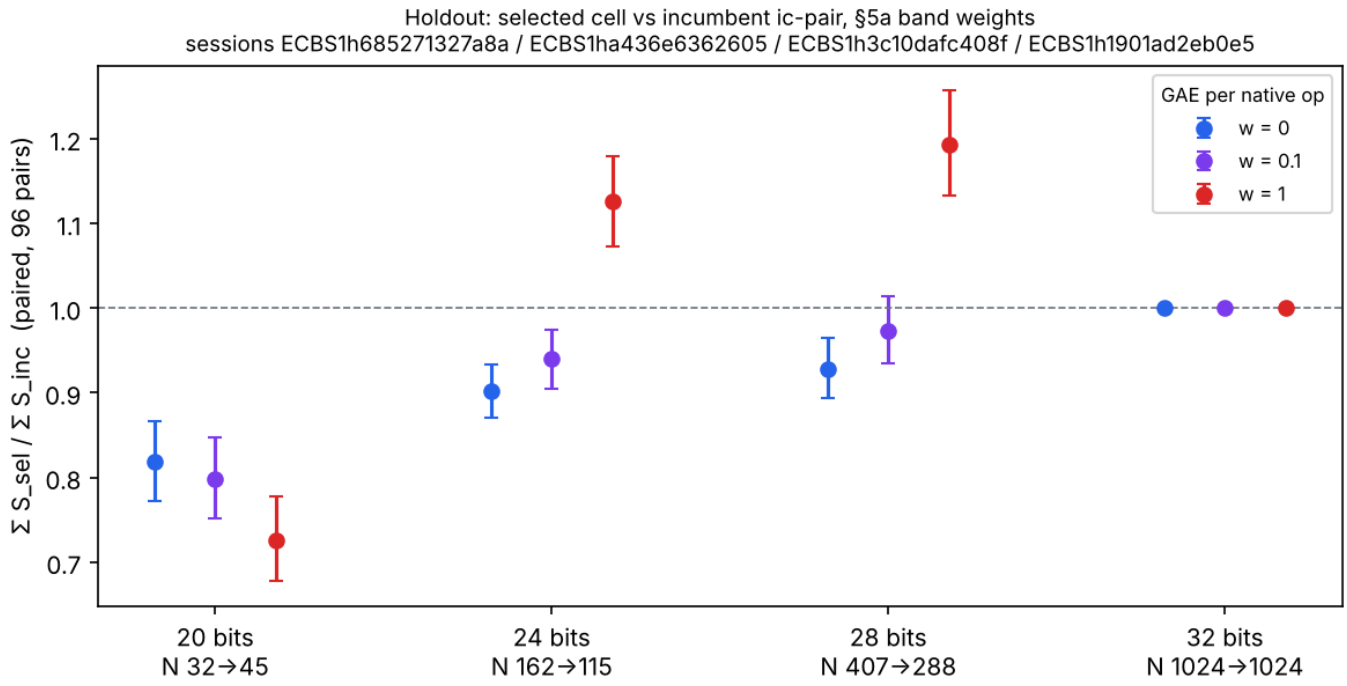


Figure 1. Holdout `Σ S_sel / Σ S_inc` per size at w = 0, 0.1 and 1, with cluster-bootstrap 95% intervals. There are 96 pairs per point. Sessions: `ECBS1h685271327a8a`, `ECBS1ha436e6362605`, `ECBS1h3c10dafc408f` and `ECBS1h1901ad2eb0e5`.

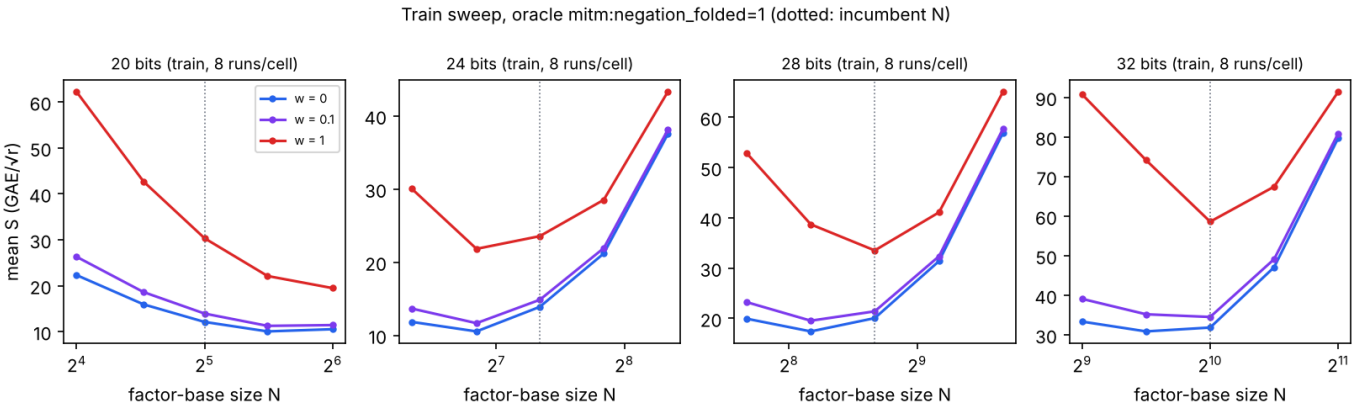


Figure 2. Train mean S against N for the incumbent oracle, 8 runs per cell. Sessions: `ECBS1h9fcb65b4770e`, `ECBS1hc463af97cc77`, `ECBS1had86361014ec` and `ECBS1h5cd4a4b40770`.

Controls

- **A/A arms:** `inc-aa` and `rho-neg-aa` give exactly 1.000 at every weight and size.

- **Matched rho:** `rho-neg`, in the same sessions.

- **Stride control:** `prime-abscissa:size=N*,stride=7919`.

- **Panel K:** `kummer` vs `lowx`, descriptive only.

- **Random-base control:** absent (PROTOCOL.md §12.2).

Raw records

`sessions/*/` (records.jsonl, session.json and receipts) are kept in the artifact store, with `.artifacts.json` pointers committed. Audit receipts are in `audits/`, and the replay receipts from verify are in `verify/`.